Continuous-Time Latent Dynamics for Long-Horizon Chaotic Forecasting under Irregular Sampling

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Abstract

World models deployed on physical systems must forecast from observations that arrive at irregular times, yet most sequence architectures update on a fixed step. We study whether continuous-time latent dynamics improve long-horizon forecasting of chaotic systems under irregular sampling when all models receive identical timing information. We adapt SynechismV23, a latent neural ODE with an elastic attractor term, golden-ratio sub-step scheduling and a laminar bypass, to integrate its latent state to arbitrary query times. We compare it with a Transformer using continuous-time positional encodings, a two-layer LSTM conditioned on each inter-observation gap, and a Mamba-style selective state-space model, on the Lorenz-63 system observed under four sampling regimes with equal mean rate and coefficients of variation from 0 to 1. Performance is measured by valid prediction time (VPT) in Lyapunov times over 30 held-out forecasts and 5 seeds, with all models trained to convergence. SynechismV23 attains the longest VPT in every regime (2.03 Lyapunov times under exponential sampling, against 1.78 for the LSTM, 0.55 for the Transformer and 0.50 for Mamba). Its advantage over the Transformer and Mamba holds in every seed (paired Wilcoxon $p < 10^{-12}$); its advantage over the LSTM under irregular sampling is consistent (4 of 5 seeds per regime) but modest (9–14%). The Transformer and Mamba lose approximately 80% of their VPT between regular and exponential sampling. We also report how the comparison changes with training budget and release all code and per-forecast results.

1 Introduction

Learned world models are increasingly used to predict the evolution of physical systems for control, robotics and immersive sensing. Two properties of such systems make this hard. First, many are chaotic: a small state error grows exponentially at a rate set by the

leading Lyapunov exponent, so every free-running forecast has a finite valid horizon. Second, observations rarely arrive on a fixed clock. Sensors drop readings, frame rates vary, communication links fail and event-driven measurements arrive in bursts.

Dominant sequence architectures, including Transformers (Vaswani et al., 2017), recurrent networks such as the LSTM (Hochreiter and Schmidhuber, 1997) and selective state-space models (Gu and Dao, 2023), update their internal state once per observation. Irregular timing can be supplied as an additional input, but the update itself remains indexed by step rather than by time. Latent ODE models (Chen et al., 2018; Rubanova et al., 2019) instead define the evolution of a latent state through a learned vector field, which can be integrated over any interval.

Latent ODEs have mainly been evaluated on interpolation, classification and short-range extrapolation of irregular series. Chaotic forecasting benchmarks, in turn, almost always assume regular sampling (Pathak et al., 2018; Vlachas et al., 2020; Gilpin, 2023). It therefore remains unclear whether continuous-time latent dynamics confer an advantage for long-horizon forecasting of chaotic systems as sampling becomes irregular, when competing architectures receive the same timing information.

Contributions.

1. A controlled benchmark that isolates sampling irregularity: Lorenz-63 observed under four regimes with equal mean rate and coefficients of variation of 0, 0.29, 0.70 and 1.0.
2. A time-fair protocol in which every model receives the past inter-observation gaps and the future query times.
3. An adaptation of the SynechismV23 latent ODE that integrates to arbitrary query times with golden-ratio sub-step scheduling.
4. A comparison against Transformer, LSTM and Mamba forecasters trained to convergence over 5 seeds, with paired statistical tests, a training-budget analysis and component ablations.

2 Related Work

Continuous-time latent models. Neural ODEs parameterize the derivative of a hidden state with a neural network (Chen et al., 2018). Latent ODEs combine this with a variational encoder to model irregularly sampled series (Rubanova et al., 2019); neural controlled differential equations (Kidger et al., 2020) and continuous recurrent units

(Schirmer et al., 2022) extend continuous-time modelling to streaming and stochastic settings. These models are typically evaluated on interpolation, classification and short-horizon extrapolation rather than long free-running forecasts of chaotic dynamics.

Discrete models with time information. GRU-D introduces learned decay over observation gaps in a recurrent unit (Che et al., 2018). Multi-time attention networks provide continuous-time embeddings for attention (Shukla and Marlin, 2021), and ContiFormer incorporates ODE dynamics into the attention mechanism (Chen et al., 2023). Selective state-space models such as Mamba use an input-dependent discretization step (Gu and Dao, 2023), which can in principle absorb uneven gaps.

Forecasting chaotic systems. Reservoir computing and recurrent networks have been evaluated on Lorenz and Kuramoto–Sivashinsky systems using valid prediction time in Lyapunov units (Pathak et al., 2018; Vlachas et al., 2020). Gilpin (2023) compares a wide range of forecasters across many chaotic systems under regular sampling. Closest to our setting, Krach and Teichmann (2024) learn a chaotic double pendulum from irregular and incomplete observations with neural jump ODEs and improve long-term prediction, but do not compare against Transformer, recurrent or state-space baselines or vary the degree of irregularity.

Benchmarks for irregular forecasting. BITS (Yan et al., 2026) evaluates many irregular-series forecasters on eleven real-world datasets and finds that differential-equation models often trail Transformer-style and transformation-based methods. Li et al. (2026) show that mean squared error on irregular data is biased by each sample's timestamp distribution. Our metric is defined in elapsed physical time (Section 4.3), which limits, but does not remove, this concern.

To our knowledge, no prior study compares continuous-time and discrete-sequence forecasters on long-horizon chaotic forecasting across graded levels of sampling irregularity, with equal timing information and reported convergence.

3 Method

3.1 Problem formulation

A trajectory is observed at increasing times $t_1 < t_2 < \dots$ with values $x_i \in \mathbb{R}^3$ and gaps $\Delta_i = t_i - t_{i-1}$. Given the last $S = 50$ observations and their gaps, (x_i, Δ_i) for $i = n-S+1, \dots, n$, and the next $K = 20$ query times $t_{n+1}, \dots, t_{n+K}$, a forecaster predicts $\hat{x}_{n+1}, \dots, \hat{x}_{n+K}$. All gaps are divided by the mean gap $\bar{\Delta} = 0.02$ before entering any network.

3.2 SynechismV23 for irregular timing

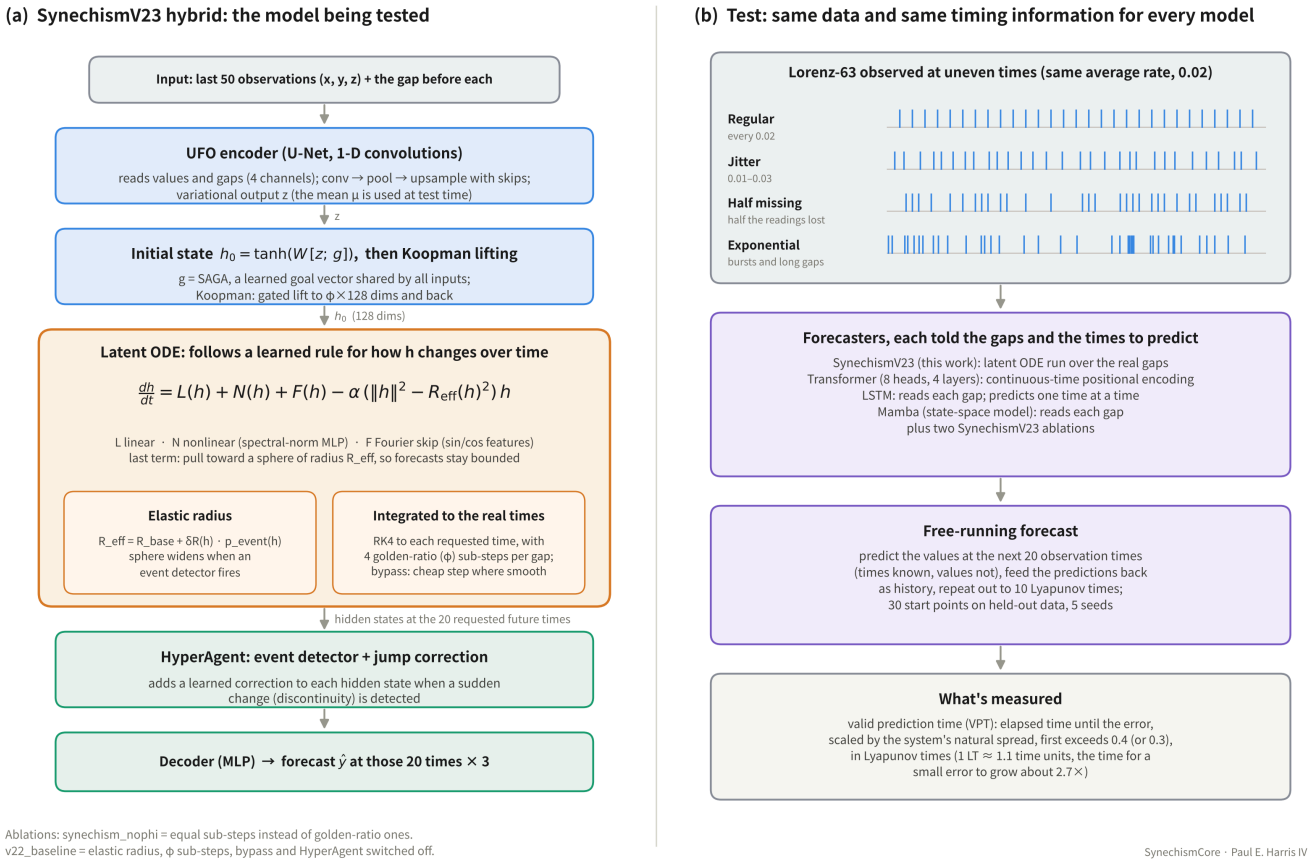


Figure 1. (a) SynechismV23 adapted to irregular sampling. (b) Evaluation protocol shared by all forecasters; tick rows show 0.6 time units of sampling times drawn from each regime.

Encoder. A one-dimensional convolutional U-Net reads the history as four channels (three state variables and the normalized gap) and outputs the mean μ and log-variance of a latent code z. During training z is sampled; at evaluation z = μ, so forecasts are deterministic.

Initial state. The code is concatenated with a learned goal vector g (SAGA) and mapped to the initial latent state, $h_0 = K(\tanh(W[z; g]))$, where K is a gated Koopman lifting that projects h into a space of dimension $\lfloor \varphi \cdot 128 \rfloor$ and back, $K(h) = \sigma(W_g h) \odot P(\mathrm{GELU}(L h)) + (1 - \sigma(W_g h)) \odot h$. The latent dimension is 128.

Latent dynamics. The latent state follows an elastic attractor ODE,

$$dh/dt = L(h) + N(h) + F(h) - \alpha(\|h\|^2 - R_eff(h)^2)\,h, \quad (1)$$

where L is linear, N is a spectrally normalized MLP, F is a Fourier-feature skip term and α = 0.1. The last term pulls the state toward a sphere of radius $R_{eff}(h) = R_{base} + \delta R(h) \cdot p(h)$, where p(h) ∈ [0, 1] is a learned event probability, so the admissible region widens during abrupt changes.

Integration to query times. Starting from h_0 at time t_n, the state is advanced to each query time in turn. Each gap Δ is divided into four sub-steps with boundaries at the sorted fractional parts {kφ mod 1}, k = 1, 2, 3, followed by 1, that is at fractions 0.236, 0.618, 0.854 and 1.0 of the gap, and each sub-step uses classical fourth-order Runge–Kutta. A laminar bypass replaces a sub-step by the gated linear update $h \leftarrow \gamma(h)(h + \delta t \cdot B\,h) + (1 - \gamma(h))\,h$ for samples whose local curvature, Var(dh/dt)/‖h‖, falls below 0.05.

Output. A HyperAgent adds a correction $p(h)\cdot[J(h) + C(h)]$ to each latent state, with jump and smooth-correction networks J and C, and an MLP decoder maps each state to $\hat{x}$.

3.3 Training objective

The model is trained on mean squared error over the K query times, plus a KL term on the latent code with weight $\beta = 5{\cdot}10^{-4}$ annealed linearly over the first 30 epochs, plus the HyperAgent regularizer on h_0.

3.4 Ablations

synechism_nophi replaces golden-ratio sub-step boundaries with four equal sub-steps. *v22_baseline* additionally removes the elastic radius, laminar bypass and HyperAgent, using a fixed-radius attractor term.

4 Experimental Setup

4.1 Data and sampling regimes

Lorenz-63 (σ = 10, ρ = 28, β = 8/3) is integrated with DOP853 (relative tolerance 10^{-10}, absolute tolerance 10^{-12}) after a warm-up of 40 time units, and the dense solution is evaluated at sampling times drawn from one of four regimes (Table 1). All regimes share a mean gap of 0.02. Each seed uses a training trajectory of 6,000 observations and an independent test trajectory. Values are standardized with training statistics. Training windows (S = 50, K = 20) are taken with stride 3.

Table 1. Sampling regimes.

Regime	Gap distribution	Coefficient of variation
Regular	$\Delta = 0.02$	0
Jitter	$\Delta = 0.02 \cdot U(0.5, 1.5)$	0.29
Missing	0.01 grid, each point retained with probability 0.5	0.70
Exponential	$\Delta = \max(0.02 \cdot \mathrm{Exp}(1), 0.002)$	1.0

4.2 Baselines

All baselines receive the normalized gap as an input feature alongside each observation, and the future gaps when forecasting (Table 2). Hidden width is 128 throughout.

Table 2. Forecasters.

Model	Use of timing information	Decoding	Parameters
SynechismV23	Gap channel in encoder; ODE integrated over true gaps	One latent state per query time	876,502
Transformer (pre-norm, 8 heads, 4 layers)	Gap feature; sinusoidal encoding of continuous time	All K outputs from the final token, conditioned on future gaps	847,292
LSTM (2 layers)	Gap feature	Autoregressive, conditioned on each next gap	201,091
Mamba-style selective SSM	Gap feature	All K outputs from the final state, conditioned on future gaps	228,044

4.3 Evaluation

From 30 start points evenly spaced along the test trajectory, each model forecasts the next K query times, appends its predictions to the history and repeats until a horizon of 10 Lyapunov times. The normalized error at query time t_k is $e(t_k) = \|(\hat{x}_k - x_k)/s\|_2 / \sqrt{3}$, where s

is the per-variable standard deviation of the standardized training data. The valid prediction time at threshold θ is

$$VPT_\theta = \lambda_{max} \cdot \min\{ t_k - t_n : e(t_k) > \theta \}, \quad (2)$$

with λ_max = 0.906, so VPT is measured in Lyapunov times (1 LT ≈ 1.10 time units). We report θ = 0.4 and θ = 0.3. Forecasts that never exceed θ within the horizon would be censored at the horizon; none were. For each seed we take the median VPT over the 30 starts, and report the mean and range of these medians over 5 seeds. Pairwise comparisons use the Wilcoxon signed-rank test on the 150 paired forecasts (5 seeds × 30 starts) evaluated on identical test windows.

4.4 Training

All models are trained with Adam, cosine learning-rate decay, batch size 64, gradient-norm clipping at 1.0 and mean squared error, with learning rates of 10^{-3} (SynechismV23 variants and LSTM), $6{\cdot}10^{-4}$ (Transformer) and $5{\cdot}10^{-4}$ (Mamba). The main comparison trains every model for 200 epochs; a preliminary comparison used 50 epochs. Per-epoch training losses are recorded for every run. Experiments ran on a single NVIDIA A100 (40 GB); one 200-epoch SynechismV23 run takes about 2 hours, and each baseline run a few minutes.

5 Results

5.1 Main comparison at convergence

With all models trained for 200 epochs, SynechismV23 attains the longest VPT in all four regimes (Figure 2, Table 3).

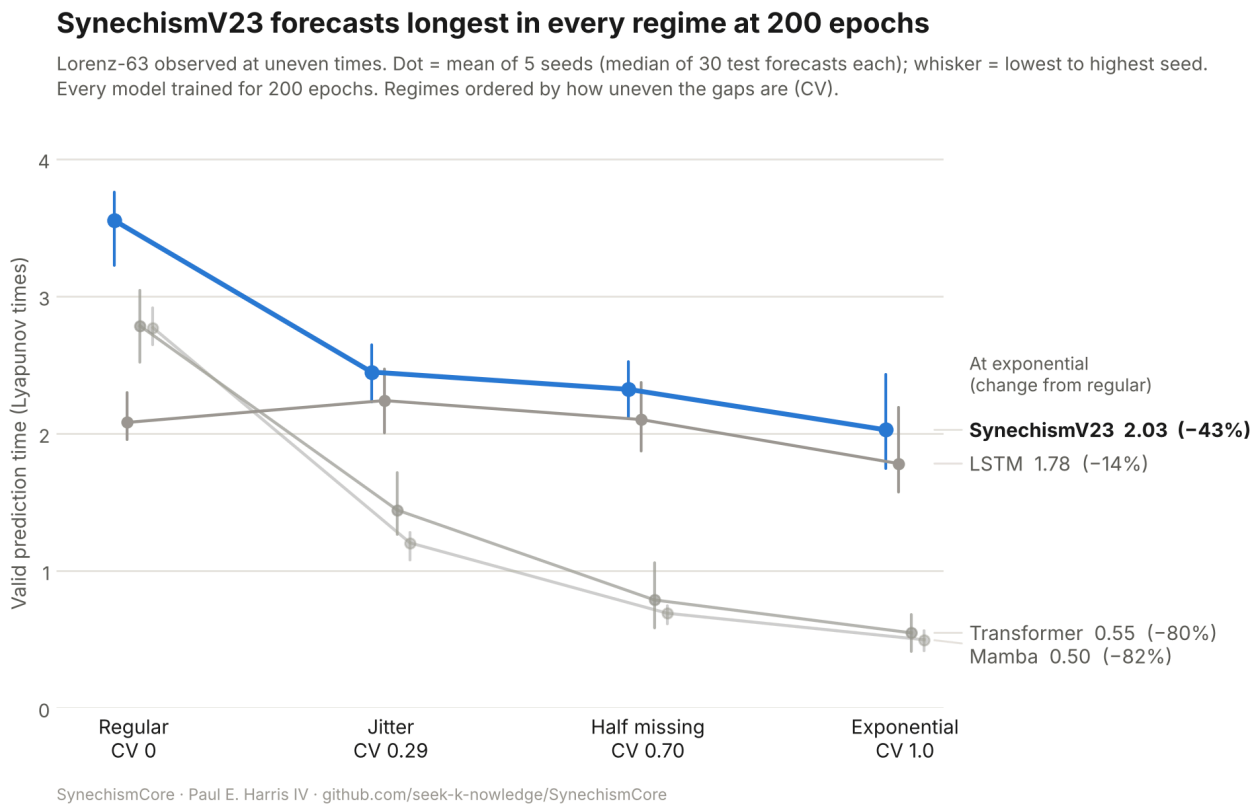


Figure 2. Mean $VPT_{0.4}$ over 5 seeds (each the median of 30 forecasts); whiskers show the lowest and highest seed. Labels give the value under exponential sampling and the relative change from regular sampling.

Table 3. $VPT_{0.4}$ in Lyapunov times, mean over 5 seeds with range in parentheses; 200 epochs.

Model	Regular	Jitter	Missing	Exponential
SynechismV23	**3.55** (3.23–3.76)	**2.45** (2.23–2.65)	**2.32** (2.12–2.53)	**2.03** (1.75–2.43)
LSTM	2.08 (1.96–2.30)	2.24 (2.01–2.47)	2.10 (1.88–2.37)	1.78 (1.57–2.19)
Transformer	2.79 (2.52–3.04)	1.44 (1.27–1.72)	0.79 (0.58–1.06)	0.55 (0.42–0.68)
Mamba	2.77 (2.65–2.92)	1.20 (1.08–1.28)	0.69 (0.62–0.75)	0.50 (0.42–0.57)

Table 4. Pairwise comparison of SynechismV23 against each baseline: seeds with higher median VPT out of 5, forecasts won out of 150, and two-sided Wilcoxon signed-rank p-value.

SynechismV23 vs	Regular	Jitter	Missing	Exponential
LSTM	$5/5 \cdot 139/150 \cdot p = 6{\cdot}10^{-21}$	$4/5 \cdot 88/150 \cdot p = 6{\cdot}10^{-4}$	$4/5 \cdot 86/150 \cdot p = 0.005$	$4/5 \cdot 86/150 \cdot p = 0.035$
Transformer	$5/5 \cdot 120/150 \cdot p = 2{\cdot}10^{-13}$	$5/5 \cdot 120/150 \cdot p = 3{\cdot}10^{-15}$	$5/5 \cdot 138/150 \cdot p = 2{\cdot}10^{-24}$	$5/5 \cdot 146/150 \cdot p = 7{\cdot}10^{-26}$
Mamba	$5/5 \cdot 117/150$	$5/5 \cdot 133/150$	$5/5 \cdot 143/150$	$5/5 \cdot 149/150$

Against the Transformer and Mamba, SynechismV23 wins every seed in every regime, and its advantage grows with irregularity, reaching a factor of 3.7 and 4.1 under exponential sampling. Against the LSTM, its advantage is large under regular sampling and modest under irregular sampling (9–14% higher mean VPT, 4 of 5 seeds per regime). Forecasts from one trained model are not independent, so the forecast-level p-values should be read as indicative; the seed counts are the conservative summary.

Degradation with irregularity. From regular to exponential sampling, mean VPT falls by 43% for SynechismV23, 14% for the LSTM, 80% for the Transformer and 82% for Mamba. The LSTM is therefore the least sensitive to irregularity, while SynechismV23 retains the highest absolute VPT.

5.2 Threshold sensitivity

At $\theta = 0.3$ the ranking is unchanged and the margin over the LSTM widens (Table 5).

Table 5. $\mathrm{VPT}_{0.3}$ in Lyapunov times, mean over 5 seeds; 200 epochs.

Model	Regular	Jitter	Missing	Exponential
SynechismV23	**3.40**	**2.30**	**2.10**	**1.81**
LSTM	1.82	1.84	1.76	1.41
Transformer	2.53	1.10	0.60	0.39
Mamba	2.59	0.95	0.56	0.35

5.3 Effect of training budget

At 50 epochs, training loss was still decreasing by 5–18% over the final 10 epochs for every model. At 200 epochs, the change over the final 20 epochs was at most 4.4% for SynechismV23, 3.9% for the LSTM, 6.3% for Mamba and 11.2% for the Transformer (regular regime; at most 4.6% elsewhere). Longer training improved all models under irregular sampling and the LSTM most (Table 6). The ranking was unchanged, but SynechismV23's margin over the LSTM under irregular sampling decreased from 19–31% to 9–14%. The LSTM's regular-sampling VPT decreased with longer training while its training loss continued to fall, consistent with improved one-step accuracy that did not translate into free-running stability; we did not investigate this further.

Table 6. Mean $VPT_{0.4}$ at 50 → 200 epochs.

Model	Regular	Jitter	Missing	Exponential
SynechismV23	2.97 → 3.55	1.68 → 2.45	1.62 → 2.32	1.50 → 2.03
LSTM	2.59 → 2.08	1.28 → 2.24	1.32 → 2.10	1.26 → 1.78
Transformer	2.40 → 2.79	1.08 → 1.44	0.53 → 0.79	0.37 → 0.55
Mamba	1.69 → 2.77	0.96 → 1.20	0.47 → 0.69	0.32 → 0.50

5.4 Ablations

Ablations were run at the 50-epoch budget only (Table 7). Replacing golden-ratio sub-steps with equal sub-steps produced no measurable change in VPT, with exact ties on 26–66 of 150 forecasts per regime. Removing the V23 components (v22_baseline) lowered VPT in 4 of 5 seeds in every regime, by 0.01 to 0.29 Lyapunov times on average. Whether these differences persist at convergence is untested.

Table 7. Mean $VPT_{0.4}$ of SynechismV23 variants at 50 epochs.

Variant	Regular	Jitter	Missing	Exponential
SynechismV23	2.97	1.68	1.62	1.50
Equal sub-steps (synechism_nophi)	2.93	1.62	1.63	1.51
V23 components removed (v22_baseline)	2.68	1.67	1.50	1.41

6 Discussion

Timing information alone does not close the gap. The Transformer and Mamba received the same gap information as SynechismV23, yet both lost roughly 80% of their forecast horizon between regular and exponential sampling. Supplying timing as an input feature is therefore not sufficient for these architectures to maintain long-horizon accuracy under irregular sampling in this setting.

Per-query conditioning appears important. The two models that remained robust to irregularity, SynechismV23 and the LSTM, both produce each forecast by advancing a state to a specific query time. The Transformer and Mamba heads emit all K outputs jointly from a single summary vector. Our results do not separate the contribution of continuous-time dynamics from that of per-query sequential decoding; variants of the Transformer and Mamba with autoregressive, gap-conditioned decoding would test this directly.

Absolute accuracy versus robustness. SynechismV23 attains the highest VPT in every regime, but the LSTM is the least affected by irregularity. Under regular sampling the converged LSTM performs worse than at 50 epochs, so part of its small relative degradation reflects a lower regular-sampling baseline.

Relation to real-data benchmarks. BITS reports that differential-equation models often trail Transformer-style models on real irregular datasets (Yan et al., 2026). The settings differ: BITS evaluates short-horizon accuracy on heterogeneous real data, while we evaluate long free-running forecasts of a known chaotic system. Whether our findings transfer to real data is an open question.

Training budget. Margins measured at an unconverged budget differed substantially from those at convergence, and the direction of change differed between models. Comparisons of forecasting architectures should report training curves alongside results.

7 Limitations and Future Work

The study uses a single three-dimensional chaotic system, a single mean sampling rate and 5 seeds per configuration; the advantage over the LSTM under irregular sampling is consistent but small relative to seed-to-seed variation. Hyperparameters were not tuned per model beyond the learning rates listed. Stronger continuous-time baselines, notably ContiFormer (Chen et al., 2023) and neural jump ODEs (Krach and Teichmann, 2024), were not included. Ablations were run only at the unconverged 50-epoch budget. VPT

measures the horizon of pointwise accuracy and does not assess whether long free-running trajectories reproduce the attractor's invariant statistics, which is where aperiodic sub-step scheduling may be most relevant.

Future work will (i) add seeds for the SynechismV23–LSTM comparison and repeat the ablations at convergence; (ii) evaluate per-query decoding variants of the Transformer and Mamba and add ContiFormer; (iii) measure long-run attractor statistics and generalization to sampling rates unseen during training; and (iv) extend the benchmark to the Kuramoto–Sivashinsky equation and to real robotic sensor data.

8 Conclusion

On Lorenz-63 observed under graded levels of sampling irregularity, a continuous-time latent ODE trained to convergence achieved the longest valid prediction time in every regime when all models received identical timing information. Transformer and selective state-space forecasters degraded sharply with irregularity despite access to the gaps, while a gap-conditioned LSTM remained competitive. These results support continuous-time latent dynamics as a strong candidate for world models that must forecast from irregularly timed observations, and motivate the controlled extensions outlined in Section 7.

Code and Data Availability

Code, configuration and per-forecast results, including the full per-epoch training losses for every run, are available at github.com/seek-k-nowledge/SynechismCore under the MIT License. The experiment is documented in `docs/IRREGULAR_TIMING.md`; results are in `results/irregular_200ep/` and `results/irregular_50ep/`.

Use of AI Tools

This work was conducted by an independent researcher with the assistance of AI tools. Claude (Anthropic) assisted with implementation, experiment orchestration, statistical analysis, figure preparation and drafting of the text. The author defined the research direction and model architecture, made the methodological decisions, including retraining all models to convergence before reporting, and reviewed all results. All reported numbers are computed from the released result files.

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